

US LNG Exports, Global Gas Prices, and Domestic Market Insulation

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Introduction and Background

The US natural gas market has changed sharply since the mid 2010s as liquefied natural gas exports connected Henry Hub more directly with global gas markets. Before large scale LNG exports, domestic prices were mainly shaped by US production, storage, weather, pipeline demand, and seasonal consumption. Since the first major Lower 48 LNG export cargo left Sabine Pass in February 2016, LNG exports have expanded rapidly. US LNG exports increased from about 0.5 billion cubic feet per day in 2016 to 15.0 billion cubic feet per day in 2025, and the US became the world's largest LNG exporter by 2024, exporting about 11.9 billion cubic feet per day (EIA, 2025, 2026).

This market change raises an important question: do rising LNG exports place upward pressure on domestic US natural gas prices, or does strong domestic production limit that effect? The question matters because Henry Hub is the main US natural gas benchmark, affecting households, power producers, industrial users, and LNG exporters. LNG exports can also support producer revenue and improve energy security for allied countries. The policy importance of this issue became clear in January 2024, when the US Department of Energy announced a pause in new non free trade agreement LNG export authorization decisions while updating its economic and environmental public interest analysis (DOE, 2024).

The structure and operation of the US gas market explain why the price effect of LNG exports is not simple. Natural gas is produced from shale and conventional fields, transported through pipelines, stored underground, consumed domestically, and increasingly exported as LNG. Henry Hub reflects domestic supply and demand conditions and is widely used as the main reference price for US gas contracts and market analysis (EIA, 2026). LNG exports allow US gas to move beyond the domestic pipeline system into Europe and Asia, where import dependence and different supply conditions often create higher prices.

The US LNG market operates through a chain in which producers sell gas into the pipeline system, LNG terminals receive gas near pipeline hubs, liquefy it, and ship it to foreign buyers when international prices justify liquefaction and transport costs. Henry Hub remains the key domestic price point, while Europe and Asia prices reflect import dependent markets where LNG cargoes can earn higher returns. Regulation matters because LNG exports to non free trade agreement countries require DOE's public interest approval, allowing federal policy to shape future export capacity even when current shipments continue.

Export growth, however, does not automatically mean higher domestic prices. If exports rise while pro-

duction and storage are tight, domestic prices may increase; if exports rise while production also expands, the price effect may be smaller. This is why dry natural gas production and storage are key controls in the analysis. Storage is especially important because natural gas demand is seasonal, with inventories usually building during lower demand months and falling during colder months.

The global energy crisis after 2021 made this relationship more important. European and Asian gas prices rose sharply relative to Henry Hub, creating large international price spreads and stronger incentives for US LNG exports. The EIA reported that LNG exports continued to drive growth in total US natural gas exports in 2023, and the US supplied nearly half of Europe's LNG imports that year (EIA, 2024). This suggests that Europe has become a major demand center for US LNG and may be an important global price signal for Henry Hub. This report examines whether rising US LNG exports have made Henry Hub prices more sensitive to global gas market conditions.

Data and Methodology

This study uses monthly data from January 2001 to January 2026 to examine whether rising US LNG exports are associated with changes in domestic natural gas prices. The dependent variable is the Henry Hub natural gas spot price, measured in dollars per MMBtu, and the main explanatory variable is US LNG exports, measured in million cubic feet. The main controls are US dry natural gas production, US natural gas storage, European and Asian benchmark prices, and monthly seasonality, because production captures domestic supply, storage captures inventory tightness, and foreign benchmarks capture global gas market conditions. The analysis first uses graphs, correlations, and period averages across five market periods: before LNG expansion, 2001 to 2015; early LNG expansion, 2016 to 2019; COVID and recovery, 2020 to 2021; global energy crisis, 2022 to 2023; and mature LNG export period, 2024 to January 2026.

The main regression model is estimated in log form because the log specification gives a clearer percentage based interpretation and allows the coefficients to be interpreted as elasticities. This is useful because LNG exports, production, storage, and benchmark prices are measured in different units and changed substantially over the sample period. The main log regression model is:

$$\log(HH_t) = \beta_0 + \beta_1 \log(LNG_t) + \beta_2 \log(PROD_t) + \beta_3 \log(STOR_t) + \beta_4 \log(EU_t) + \beta_5 \log(ASIA_t) + Month_t + \varepsilon_t$$

where HH_t is Henry Hub price, LNG_t is US LNG exports, $PROD_t$ is US dry natural gas production, $STOR_t$ is US natural gas storage, EU_t is the European benchmark price, $ASIA_t$ is the Asian benchmark

price, and $Month_t$ controls for monthly seasonality. The main coefficient of interest is β_1 , which shows the percentage change in Henry Hub price associated with a 1% change in LNG exports, holding domestic supply, storage, global prices, and seasonality constant.

Because monthly energy market variables may trend over time, Augmented Dickey Fuller tests are used to check whether the logged variables are stationary or nonstationary. This step is important because regressions using trending time series can produce misleading relationships if the variables do not share a meaningful long run connection. First difference ADF tests are also used to check whether nonstationary variables become stationary after taking monthly changes. Engle Granger cointegration tests are then used to examine whether Henry Hub prices, LNG exports, and global benchmark prices share a long run relationship. This is done by testing whether the residuals from long run log regressions are stationary. If the residuals are stationary, the variables are cointegrated and the log level regression can be interpreted as a long run relationship rather than a spurious time series regression.

A level regression is still reported as a supporting specification, while a first difference model examines short run monthly changes. Post 2016 and post 2022 interaction models test whether the export price relationship changed after major market events, and Granger causality and lagged regressions provide additional predictive checks. Newey West standard errors are used because monthly energy market data may have autocorrelation across nearby months and heteroskedasticity during volatile periods, making the significance tests more reliable without changing the regression coefficients.

The analysis also includes a liquefaction cost and export margin extension. Since LNG exports do not connect Henry Hub to Europe and Asia automatically, an implied LNG cost proxy is calculated as $1.15 \times HH + 3$. Europe and Asia export margin proxies are then calculated by subtracting this implied LNG cost from European and Asian benchmark prices. These margin measures help show when global prices are high enough to create stronger export incentives and transmit pressure back toward the US market (Corpus Christi Liquefaction, 2017).

Results and Discussion

The graphical results show a clear shift in the US natural gas market. Figure A1 shows that Henry Hub prices were volatile in the early 2000s, declined after the shale gas expansion, and rose again during the 2021 to 2022 global energy shock. Figure A2 shows a different pattern for LNG exports, which remained low before 2016 and then increased sharply as the US became a major LNG exporter. This creates the central puzzle of the analysis: LNG exports expanded strongly, but Henry Hub prices did not rise in the same way.

Figures A3 and A4 help explain this puzzle. European and Asian gas prices became much more volatile

than Henry Hub during the global energy crisis, and the spreads between foreign benchmarks and Henry Hub widened sharply. These spreads created strong incentives for US LNG exports, but the smaller domestic price response shows that the US market did not fully absorb global price volatility. LNG exports increased the connection between the US and global markets, but export capacity, liquefaction limits, transportation costs, storage, and domestic supply conditions prevented full price equalization.

Domestic supply is central to this result. Figure A5 shows strong growth in US dry natural gas production, while Figure A6 shows the seasonal role of storage in balancing demand across the year. Table A1 confirms this pattern more clearly. Before LNG expansion, average LNG exports were about 3,981 million cubic feet and average Henry Hub price was about \$5.06 per MMBtu. During early LNG expansion, average LNG exports rose to about 79,105 million cubic feet, but Henry Hub fell to about \$2.79. This happened because average production also increased from about 1.77 trillion cubic feet per month to 2.47 trillion cubic feet per month. Even in the mature LNG export period, average exports reached their highest level while Henry Hub averaged only about \$3.07, supported by production above 3.21 trillion cubic feet per month.

The correlation plot in Figure A7 shows why simple relationships can be misleading. LNG exports are strongly correlated with broader export and production variables, meaning the export boom occurred alongside broader supply growth. The raw correlation between LNG exports and Henry Hub price is negative, but this does not mean LNG exports lower domestic prices. It mainly reflects that exports increased during a period when production growth placed downward pressure on Henry Hub.

The main log regression results in Table A3 provide the strongest evidence for the central relationship. After controlling for production, storage, European price, Asian price, and monthly seasonality, LNG exports have a positive and statistically significant coefficient. A 1% increase in LNG exports is associated with about a 0.049% increase in Henry Hub price, holding other factors constant. This means LNG exports are associated with some upward pressure on Henry Hub prices, but the effect is economically modest. Production and storage have much larger negative coefficients, showing domestic supply fundamentals strongly reduce domestic price pressure. Europe price is positive and significant, while Asia price is not significant as Europe price is included, suggesting that Europe is the stronger global price signal in this sample.

Table A13 provides the ADF and cointegration test results. The ADF tests show that some logged variables are stationary in levels, including Henry Hub price and storage, while LNG exports, production, European prices, and Asian prices are nonstationary in levels. After first differencing, all variables become stationary. This means the trending variables become stable after taking monthly changes, which supports using first differences for short run analysis. The Engle Granger cointegration results show only weak evidence of cointegration between Henry Hub and LNG exports alone, since the residual p value is 0.051. However,

Henry Hub and European gas prices are cointegrated, with a residual p value of 0.015. Most importantly, the full log model residual is stationary, with a p value of 0.010. This supports the use of the full log level regression as a long run relationship model rather than a purely spurious time series regression.

The short run results are more cautious. Table A4 shows that monthly changes in LNG exports do not significantly explain monthly changes in Henry Hub prices, while monthly changes in Europe prices are positive and significant. This suggests that LNG exports matter more as a long run market integration channel than as a direct month to month driver. Foreign prices still affect Henry Hub through expectations, export incentives, contract behavior, and arbitrage pressure, but the pass through is partial. US gas supply appears more responsive than Europe and Asia because of large production, storage, and infrastructure, but not perfectly elastic, since Henry Hub still rose in 2021 to 2022, only by less than European and Asian prices.

Tables A5 and A6 test whether the LNG export effect changed after 2016 or after 2022. The interaction terms are not statistically significant, so the data does not show a clean structural break in the export price relationship after either event. This does not mean those periods were unimportant. Rather, the events changed the market environment, while the marginal relationship between LNG exports and Henry Hub prices remained shaped by production, storage, and global benchmark prices.

Table A7 reports Granger causality tests, which examine predictive relationships rather than true structural causation. The results show no significant predictive relationship in either direction at 1, 2, or 3 month lags. Past LNG exports do not significantly predict Henry Hub prices, and past Henry Hub prices do not significantly predict LNG exports. Table A8 adds a lagged regression check, where only the 1 month LNG export lag is positive and significant, while the 2 and 3 month lags are not. Table A9 further supports the main result: after removing Asia price, LNG exports remain positive and significant, production and storage remain negative and significant, and Europe price remains strongly positive.

Tables A10 and A11 compare the log model before and after 2016. Before 2016, LNG exports are not statistically significant, while production, storage, and Europe price are significant. After 2016, LNG exports remain statistically insignificant, but Europe price remains positive and strongly significant. This result supports the broader interpretation that LNG exports alone do not explain Henry Hub movements in each subperiod, while Europe continues to appear as an important global price channel. Table A12 also shows that lagged LNG exports are more informative than same month export volume, with the 1 month and 2 month LNG export lags positive and significant in the log lagged regression.

The liquefaction margin extension adds another layer to the market integration story. Figure A9 and Table A14 show that Europe and Asia export margin proxies widened sharply during the global energy crisis and remained positive during the mature LNG export period. This suggests that global price transmission de-

depends not only on foreign benchmark prices, but also on whether those prices are high enough to cover the implied cost of exporting US LNG. Liquefaction economics therefore act as a filter between Henry Hub and global prices. Global prices create stronger pressure on the US market when export margins are large, but the connection remains partial because domestic production, storage, and liquefaction capacity still limit full pass through.

Overall, the evidence supports a market integration interpretation rather than a simple export volume explanation. LNG exports are positively associated with Henry Hub prices in the full log model after controlling for domestic production, storage, global prices, and seasonality, but the effect is limited rather than dominant. The ADF and cointegration tests strengthen this interpretation by showing that the full log model represents a long run relationship among the variables. Strong US production and storage reduce domestic price pressure, while European gas market conditions play an important role in both the level and short movement of Henry Hub prices. The US natural gas market has become more connected to global LNG demand, but it has not become fully exposed to the price volatility observed in Europe and Asia.

Policy Event Window

The event window around the January 2024 LNG approval pause shows a useful policy pattern. Figure A8 shows that US export share declined from roughly 12% to 13% in early 2024 to around 10% by mid 2024, but then recovered into early 2025. This does not suggest a lasting reduction in export activity. The pattern is consistent with the policy design because the pause applied mainly to pending and future non free trade agreement export approvals, not to LNG exports from already operating projects. Therefore, the pause appears more relevant for future export capacity and long term market development than for immediate short run export share.

The policy event also supports the broader conclusion that LNG exports affect the domestic market through capacity, expectations, and long run infrastructure rather than through a simple immediate shipment effect. Existing LNG terminals continued operating, previously contracted cargoes continued moving, and domestic production and storage conditions remained important for short run Henry Hub pricing. This helps explain why the data show only a limited immediate market response to the pause.

Conclusion

This report shows that US LNG exports have made the domestic natural gas market more connected to global markets, but their effect on Henry Hub prices is limited and conditional. LNG exports increased

sharply after 2016, yet Henry Hub prices did not rise in the same continuous pattern because US dry natural gas production and storage also expanded, helping absorb export demand. The main log regression results show that LNG exports are positively and significantly associated with Henry Hub prices after controlling for production, storage, global benchmark prices, and seasonality, but the effect is economically modest. Production and storage have stronger negative effects, while European benchmark prices are consistently important. The ADF and cointegration results support the use of the log level model as a long run relationship model because several variables are nonstationary in levels but become stationary after first differencing, and the full log model residual is stationary. This means the main log model is not simply capturing unrelated trends. At the same time, the first difference and Granger causality results show that LNG exports should not be treated as a simple short run causal driver of monthly Henry Hub price changes. Overall, LNG exports place some upward pressure on domestic prices, but strong US supply fundamentals, storage capacity, infrastructure, and liquefaction limits reduce the pass through from global price shocks. Europe appears to be the stronger global price channel into Henry Hub, while Asia prices are less significant after Europe is included. The liquefaction margin analysis also shows that global price transmission is strongest when foreign prices exceed the implied cost of exporting US LNG. Therefore, Henry Hub is no longer purely domestic, but it is also not a fully global LNG price pass through market.

References

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Appendix

Figure A1. Henry Hub Natural Gas Price

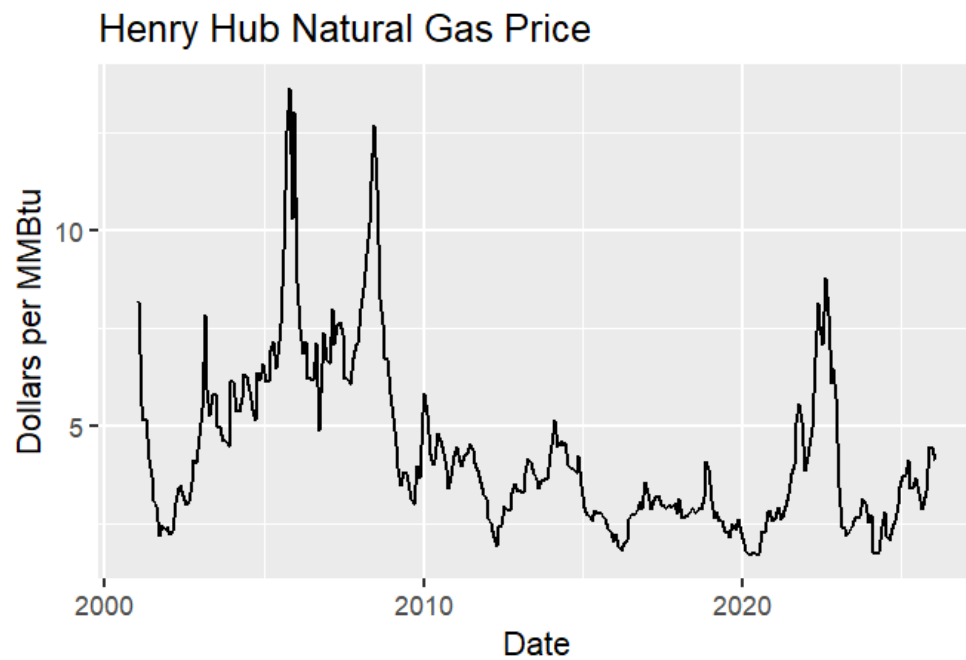


Figure A2. US LNG Exports

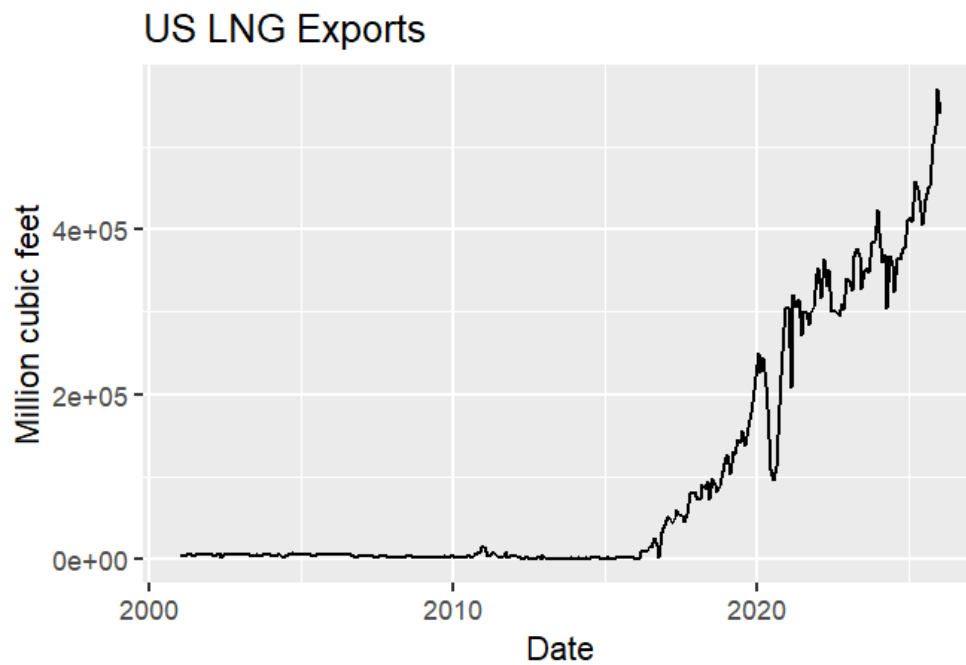


Figure A3. Natural Gas Prices across US, Europe, and Asia

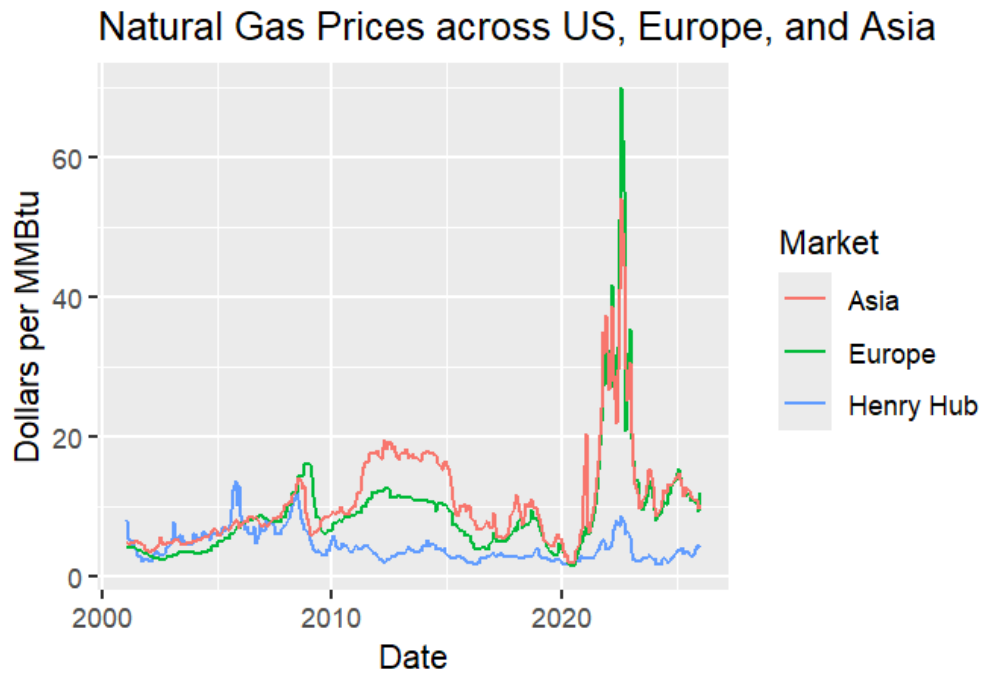


Figure A4. Global Price Spreads over Henry Hub

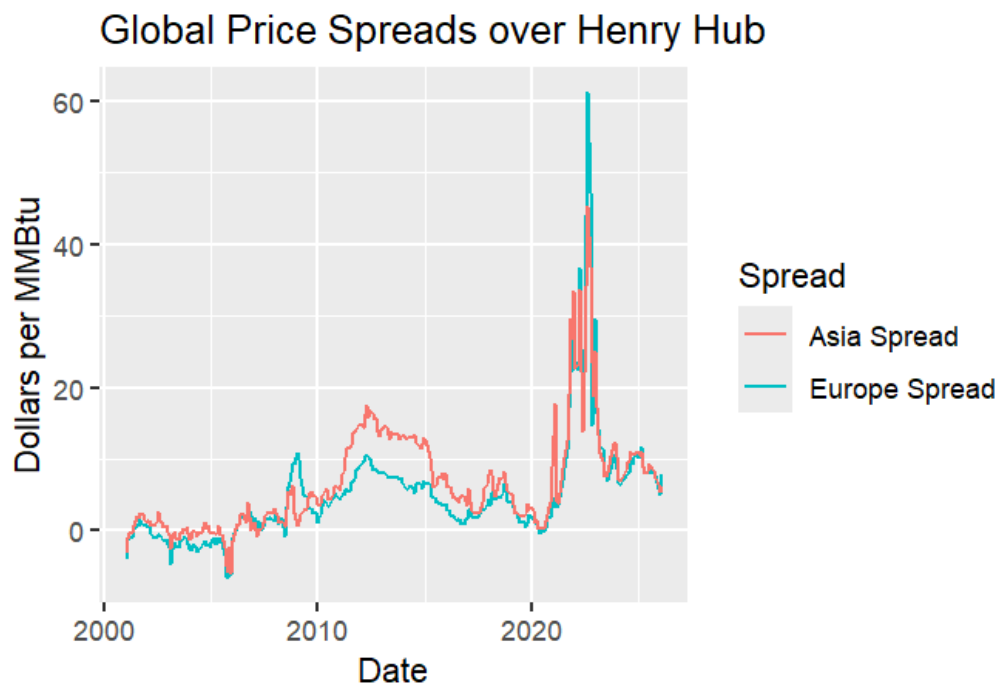


Figure A5. US Dry Natural Gas Production

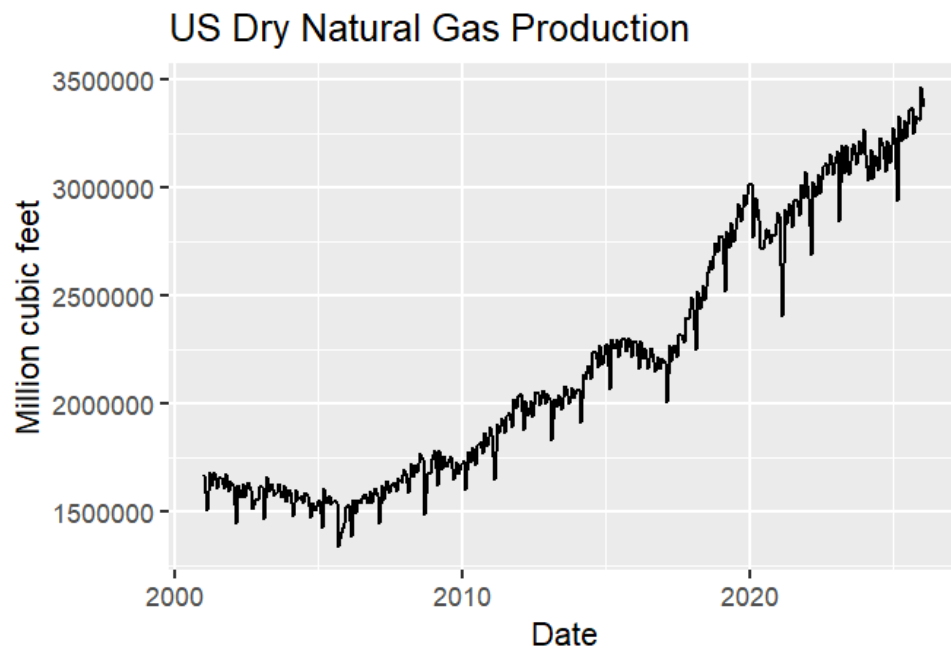


Figure A6. US Natural Gas Storage

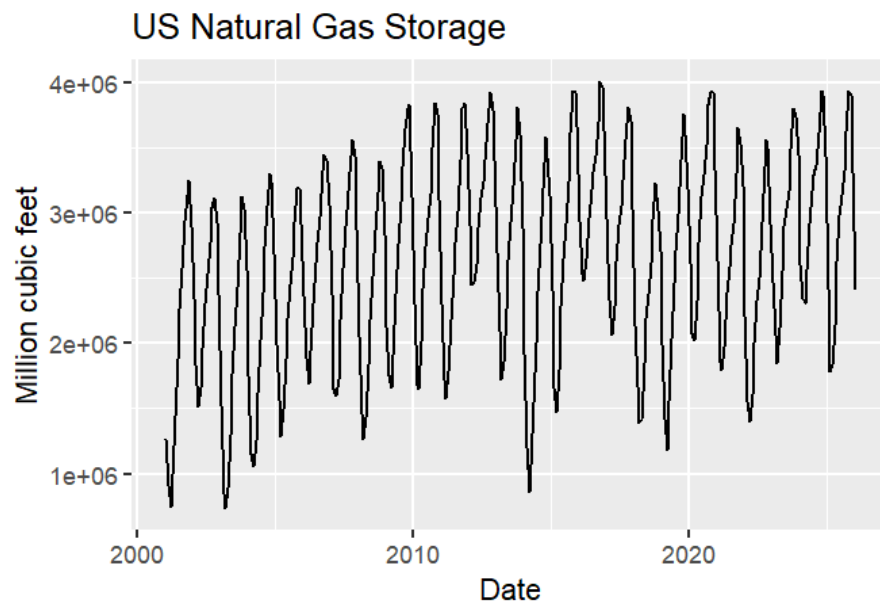


Figure A7. Correlation Plot of Main Energy Market Variables

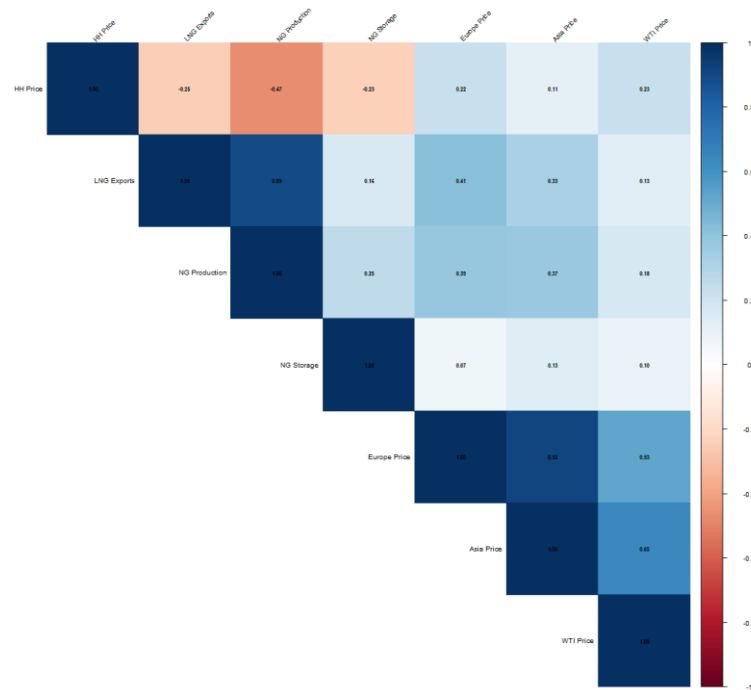


Figure A8. US Export Share around January 2024 LNG Approval Pause

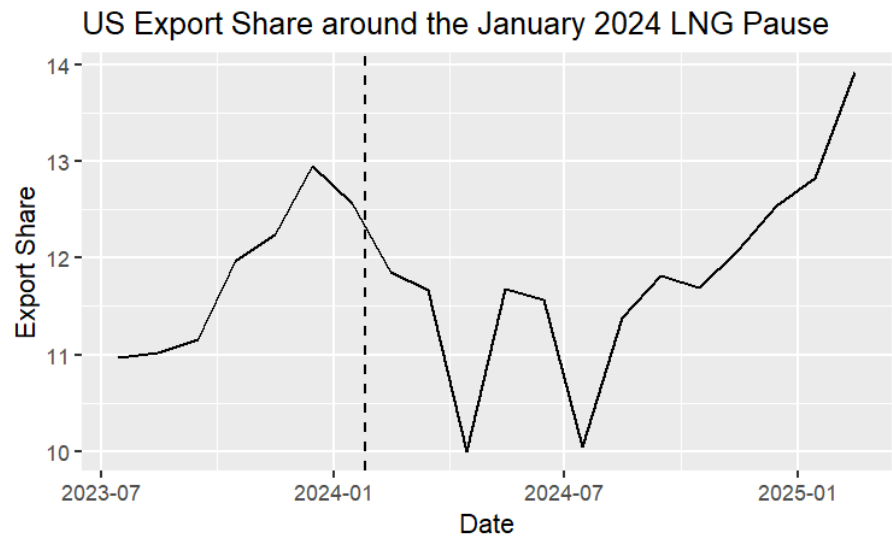


Figure A9. Europe and Asia LNG Export Margin Proxies

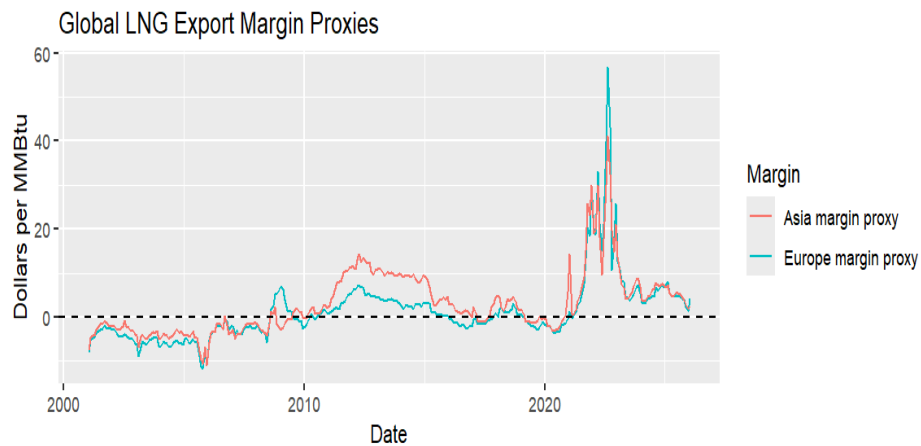


Table A1. Descriptive Statistics by Market Period

Market Period	Avg. Henry Hub Price	Avg. LNG Exports	Avg. Gas Production	Avg. Gas Storage	Avg. Europe Price	Avg. Asia Price	Avg. Europe Spread	Avg. Asia Spread
Before LNG expansion	5.06	3981.2	1768041	2511891	7.85	10.06	2.79	5.00
Early LNG expansion	2.79	79105.1	2470958	2744235	5.62	7.48	2.82	4.69
COVID and recovery	2.92	247949.3	2847517	2894078	9.54	11.48	6.62	8.56
Global energy crisis	4.59	342027.9	3079474	2672504	25.20	23.38	20.61	18.79
Mature LNG export period	3.07	416582.0	3216652	2998978	11.43	11.84	8.35	8.77

Table A2. Main Regression Results

Variable	Coefficient	Newey West SE	t value	p value
Intercept	14.8009	3.2939	4.493	0.0000
US LNG exports	8.6707e-06	3.3608e-06	2.580	0.0104
US dry natural gas production	-3.7643e-06	1.2985e-06	-2.899	0.0040
US natural gas storage	-1.4862e-06	3.9095e-07	-3.802	0.0002
Europe benchmark price	0.1541	0.0493	3.128	0.0019
Asia benchmark price	-0.0533	0.0525	-1.016	0.3104

Note: The dependent variable is Henry Hub natural gas price. Newey West standard errors are reported to account for heteroskedasticity and autocorrelation. Monthly controls are included but not reported. Observations = 301. Adjusted R squared = 0.5639.

Table A3. Log Regression Results

Variable	Coefficient	Newey West SE	t value	p value
Log LNG exports	0.0488	0.0126	3.860	0.0001
Log production	-1.3734	0.2605	-5.271	0.0000
Log storage	-0.8319	0.1564	-5.318	0.0000
Log Europe price	0.5075	0.1849	2.745	0.0064
Log Asia price	-0.1053	0.2208	-0.477	0.6338

Note: Dependent variable is log Henry Hub price. Monthly controls are included but not reported. Newey West standard errors are reported. Observations = 301. Adjusted R squared = 0.7206.

Table A4. First Difference Regression Results

Variable	Coefficient	Newey West SE	t value	p value
Change in log LNG exports	-0.0024	0.0036	-0.686	0.4933
Change in log production	-0.0067	0.1175	-0.057	0.9545
Change in log storage	0.0048	0.0603	0.080	0.9361
Change in log Europe price	0.2326	0.0487	4.778	0.0000
Change in log Asia price	0.0000	0.0551	-0.000	0.9998

Note: Dependent variable is monthly change in log Henry Hub price. Newey West standard errors are reported. Observations = 300. Adjusted R squared = 0.0392.

Table A5. Post 2016 Interaction Model Results

Variable	Coefficient	Newey West SE	t value	p value
Intercept	16.6880	3.8315	4.3555	0.0000
US LNG exports	-5.7878e-05	4.7775e-05	-1.2115	0.2267
Post 2016	0.6273	0.6417	0.9776	0.3291
US LNG exports $\times$ Post 2016	6.7632e-05	4.8801e-05	1.3859	0.1669
US dry natural gas production	-4.6942e-06	1.6536e-06	-2.8388	0.0049
US natural gas storage	-1.5427e-06	3.7064e-07	-4.1622	0.0000
Europe benchmark price	0.1388	0.0521	2.6641	0.0082
Asia benchmark price	-0.0310	0.0609	-0.5082	0.6117

Note: Dependent variable is Henry Hub natural gas price. Monthly controls are included but not reported. Newey West standard errors are reported. Observations = 301. Adjusted R squared = 0.5698.

Table A6. Post 2022 Interaction Model Results

Variable	Coefficient	Newey West SE	t value	p value
Intercept	14.7190	3.2829	4.4835	0.0000
US LNG exports	8.9463e-06	3.9667e-06	2.2554	0.0249
Post 2022	-1.3170	1.6988	-0.7752	0.4388
US LNG exports $\times$ Post 2022	2.7663e-06	4.2409e-06	0.6523	0.5147
US dry natural gas production	-3.7506e-06	1.3258e-06	-2.8291	0.0050
US natural gas storage	-1.4814e-06	4.0734e-07	-3.6367	0.0003
Europe benchmark price	0.1639	0.0572	2.8649	0.0045
Asia benchmark price	-0.0565	0.0547	-1.0333	0.3023

Note: Dependent variable is Henry Hub natural gas price. Monthly controls are included but not reported. Newey West standard errors are reported. Observations = 301. Adjusted R squared = 0.5625.

Table A7. Granger Causality Test Results

Direction tested	Lag length	F statistic	p value	Result
LNG exports $\rightarrow$ Henry Hub price	1 month	0.1622	0.6874	Not significant
LNG exports $\rightarrow$ Henry Hub price	2 months	0.1303	0.8779	Not significant
LNG exports $\rightarrow$ Henry Hub price	3 months	0.5666	0.6374	Not significant
Henry Hub price $\rightarrow$ LNG exports	1 month	0.4667	0.4951	Not significant
Henry Hub price $\rightarrow$ LNG exports	2 months	0.3259	0.7222	Not significant
Henry Hub price $\rightarrow$ LNG exports	3 months	0.3263	0.8064	Not significant

Note: Granger causality tests examine predictive relationships, not true economic causation.

Table A8. Lagged Regression Results

Variable	Coefficient	Newey West SE	t value	p value
Intercept	14.7040	3.0080	4.8885	0.0000
LNG exports lag 1	6.5627e-06	2.6901e-06	2.4396	0.0153
LNG exports lag 2	7.2157e-08	2.2534e-06	0.0320	0.9745
LNG exports lag 3	1.9071e-06	2.6417e-06	0.7219	0.4709
US dry natural gas production	-3.6726e-06	1.1666e-06	-3.1481	0.0018
US natural gas storage	-1.5064e-06	3.8448e-07	-3.9182	0.0001
Europe benchmark price	0.1529	0.0509	3.0037	0.0029
Asia benchmark price	-0.0537	0.0498	-1.0778	0.2820

Note: Dependent variable is Henry Hub natural gas price. Monthly controls are included but not reported. Newey West standard errors are reported. Adjusted R squared = 0.5542.

Table A9. Europe Focused Regression Results

Variable	Coefficient	Newey West SE	t value	p value
Intercept	15.0660	3.3019	4.5628	0.0000
US LNG exports	9.5017e-06	3.5732e-06	2.6592	0.0083
US dry natural gas production	-3.9418e-06	1.3641e-06	-2.8897	0.0042
US natural gas storage	-1.5296e-06	4.0017e-07	-3.8225	0.0002
Europe benchmark price	0.1082	0.0177	6.1273	0.0000

Note: Dependent variable is Henry Hub natural gas price. Monthly controls are included but not reported. Newey West standard errors are reported. Adjusted R squared = 0.5611.

Table A10. Log Regression Results, 2001 to 2015

Variable	Coefficient	Newey West SE	t value	p value
Log LNG exports	0.0143	0.0142	1.004	0.3169
Log production	-2.9967	0.9955	-3.010	0.0030
Log storage	-1.0625	0.2466	-4.309	0.0000
Log Europe price	0.4492	0.1263	3.557	0.0005
Log Asia price	0.2746	0.2552	1.076	0.2836

Note: Dependent variable is log Henry Hub price. Monthly controls are included but not reported. Newey West standard errors are reported. Observations = 180. Adjusted R squared = 0.7384.

Table A11. Log Regression Results, 2016 to January 2026

Variable	Coefficient	Newey West SE	t value	p value
Log LNG exports	0.0175	0.0146	1.198	0.2337
Log production	-0.8799	0.3958	-2.223	0.0284
Log storage	-0.4939	0.3486	-1.417	0.1595
Log Europe price	0.5240	0.1234	4.245	0.0000
Log Asia price	-0.1663	0.1161	-1.432	0.1552

Note: Dependent variable is log Henry Hub price. Monthly controls are included but not reported. Newey West standard errors are reported. Observations = 121. Adjusted R squared = 0.7210.

Table A12. Log Lagged Regression Results

Variable	Coefficient	Newey West SE	t value	p value
Log LNG exports lag 1	0.0198	0.0077	2.583	0.0103
Log LNG exports lag 2	0.0224	0.0075	3.010	0.0028
Log LNG exports lag 3	0.0147	0.0083	1.774	0.0772
Log production	-1.4205	0.2923	-4.860	0.0000
Log storage	-0.8540	0.1966	-4.345	0.0000
Log Europe price	0.4861	0.1824	2.665	0.0081
Log Asia price	-0.0784	0.2258	-0.347	0.7287

Note: Dependent variable is log Henry Hub price. Monthly controls are included but not reported. Newey West standard errors are reported. Observations = 298. Adjusted R squared = 0.7238.

Table A13. ADF and Cointegration Test Results

Tested series or residual	ADF statistic	p value	Result
Log Henry Hub price	-3.72	0.0233	Stationary
Log LNG exports	-1.68	0.7120	Nonstationary
Log production	-3.25	0.0799	Nonstationary
Log storage	-6.64	0.0100	Stationary
Log Europe price	-2.88	0.2050	Nonstationary
Log Asia price	-2.59	0.3270	Nonstationary
Change in log Henry Hub price	-6.79	0.0100	Stationary
Change in log LNG exports	-7.79	0.0100	Stationary
Change in log production	-8.58	0.0100	Stationary
Change in log storage	-16.63	0.0100	Stationary
Change in log Europe price	-6.43	0.0100	Stationary
Change in log Asia price	-8.53	0.0100	Stationary
Cointegration residual: Henry Hub and LNG exports	-3.42	0.0510	Not stationery at 5%
Cointegration residual: Henry Hub and Europe price	-3.89	0.0151	Stationary
Cointegration residual: full log model	-4.35	0.0100	Stationary

Table A14. Liquefaction Cost Proxy and Export Margins by Market Period

Variable	Avg HH Price	Avg Implied LNG Cost	Avg Europe Price	Avg Asia Price
Before LNG Expansion	5.06	8.82	7.85	10.1
COVID and recovery	2.92	6.36	9.54	11.5
Early LNG expansion	2.79	6.21	5.62	7.48
Global energy crisis	4.59	8.28	25.2	23.4
Mature LNG export	3.07	6.53	11.4	11.8